

Bound Git maintenance during disposable verification

LOCAL EVIDENCE / REMOTE CI NOT YET RUN

Report ID 20261002-093545-foreground-git-maintenance
Prepared UTC 2026-10-02 09:35:45
Author / role Bijan Mousavi
Parent commit: 11fc8091ced66d3b2440f7daf9650f9e08b2470c
Candidate SHA-256: 7f3dbde1f455726aef995f3e8e5dcbceaef6b09501dc20b48e6d9464738de440

Fingerprint excludes only this report's JSON/TEX/PDF; it is not the final commit SHA.

WHY / problem and value

The branch-head check failed while cleaning a temporary Git fixture; the integration check passed. Detached maintenance can outlive the checked Git command.

WHAT / change scope

Run automatic maintenance and garbage collection without detaching in epoch and adapter processes.

HOW / implementation rationale

Append command-scoped Git configuration while preserving inherited entries; reject invalid counts. Keep original historical tests and reports.

WHERE / affected paths

scripts/verification/reported_candidate.py
tests/verification/test_reported_candidate.py
docs/engineering/integration.md

PRESERVED / existing contracts

Original application, scientific content, runtime commands, locked dependencies and required checks.

RISK / what could go wrong

Foreground maintenance can increase a gate duration; existing timeouts remain enforced.

LIMITATIONS / what is not proved

The exact Linux cleanup race was not reproduced locally. A forced automatic repack completed with all traced child exits. Independent corrected head and integration CI remain unrun at authoring.

ROLLBACK / restore and verify

Use an admitted reported revert of the environment helper, tests and documentation; retain original failure and process evidence.

VALIDATION / actual local evidence

Gate	Command / check	Outcome	Duration	Evidence
Toolchain	npm run doctor	PASS	0.29 s	artifacts/admission/20261002-093545-foreground-git-maintenance/results.json
Developer tests	npm test	PASS	1.57 s	artifacts/admission/20261002-093545-foreground-git-maintenance/results.json
Report tools	npm run report-test	PASS	10.76 s	artifacts/admission/20261002-093545-foreground-git-maintenance/results.json
Staged con-trols	npm run verification-test	PASS	15.24 s	artifacts/admission/20261002-093545-foreground-git-maintenance/results.json
Index hygiene	git diff --cached --check	PASS	0.02 s	artifacts/admission/20261002-093545-foreground-git-maintenance/results.json

Remote validation at authoring: NOT_RUN. Later CI belongs to the immutable commit's check records, not a follow-up report commit.